JACKSON CMELAK

Software Engineer

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EDUCATION

B.S. Computer Science **Sept 2022 – Dec 2025**
University of California, Davis *Davis, CA*

- Relevant coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms, Database Systems, Machine Learning.

EXPERIENCE

Freelance Web Developer **Dec 2025 – Present**
Self-Employed *Laguna Niguel, CA*

- Developed and deployed production business websites in Svelte and TypeScript from specification to maintenance, integrating REST APIs and third-party SDKs with cloud data storage and hosting on Zoho Catalyst.
- Configured DNS records, TLS certificates, and MX-routed business email by hand for 2 client domains.

Data Analyst Intern **Jun 2022 – Aug 2022**
US Cryotherapy *San Ramon, CA*

- Consolidated five years of operational data from point-of-sale, marketing, and web analytics systems into one dataset, then built regression models in Python (NumPy, Pandas) to drive sales recommendations.
- Presented findings in a full-day workshop to ownership and management, who built the following year's advertising strategy using them.

PROJECTS

BoosterTutor **Python / FastAPI / PostgreSQL / SvelteKit**
AI-Powered Game Analysis App *Deployed*

- Built a grounded LLM pipeline on the Claude API: Python computes every verdict from PostgreSQL data and Claude Haiku and Sonnet only explain it from versioned prompts, so no statistic appears unless the prompt contained it.
- Built the project with an AI coding agent (Claude Code), maintaining the agent's context files, architecture decision records (ADRs) and a domain glossary so its changes stayed consistent with the design.
- Deployed the API on Render and the front end on Vercel, with per-call LLM cost logging, a daily spending cap, rate limiting, and Sentry error reporting with per-request IDs.
- Calibrated draft bots on ~1.5M human picks; with a 43,650-draft backtest fit.

Bone Fracture Detector **Python / PyTorch / scikit-learn**
Medical Image Classifier *Completed*

- Fine-tuned a ResNet with transfer learning, raising accuracy from 58.6% (hand-built features + SVM baseline) to 83.4% on the same 338-image test set, and benchmarked the results against a published study.

RapidCCNA **Godot 4.7 / GDScript / GitHub Actions**
CCNA 200-301 Study Game *In Development*

- Built a Cisco IOS CLI simulator and 14 labs graded on network behavior; 244 automated tests.
- Found answer-key bias (longest option right 58% of the time), cut it to 36%, and gated CI at 40%.

UDP Telemetry Receiver **C++17 / CMake / BSD Sockets**
Personal Project *Completed*

- Designed a fixed 28-byte binary protocol for simulated GPS/IMU data, sent between a C++17 sender and receiver.
- Hand-wrote table-driven CRC-32 and byte ordering in a CMake library shared by both endpoints.

TECHNICAL SKILLS

AI / ML: Claude API, prompt engineering, LLM output grounding, Claude Code, PyTorch, TensorFlow, scikit-learn
Languages: Python, TypeScript, C++, C, Java, SQL, Bash
Frameworks & Tools: Linux, Git, GitHub Actions, CMake, Godot, FastAPI, PostgreSQL, SvelteKit, pytest, Sentry
Networking: TCP/IP, DNS, VLANs, OSPF, UDP sockets